

Lecture 9: The Quotient Rule; Derivatives of the Other Trigonometric Functions

GOAL: Be able to differentiate functions of the form $\frac{f(x)}{g(x)}$. Use this to derive the derivatives of all 6 trigonometric functions.

Summary:

$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$		
$(\sin x)' = \cos x$	$(\cos x)' = -\sin x$	$(\tan x)' = \sec^2 x$
$(\sec x)' = \sec x \tan x$	$(\csc x)' = -\csc x \cot x$	$(\cot x)' = -\csc^2 x$

NOTE: $\left[\frac{1}{x}\right]' = -\frac{1}{x^2} \neq 0 = \frac{[1]'}{[x]}'$

$$\left[\frac{f}{g}\right]' = \left[f \cdot \frac{1}{g}\right]'$$

Q: What's $\left[\frac{1}{g}\right]'$?

A: $\left[\frac{1}{g(x)}\right]' = -\frac{g'(x)}{[g(x)]^2}$

So,
$$\begin{aligned}\left[\frac{f(x)}{g(x)}\right]' &= \left[f(x) \cdot \frac{1}{g(x)}\right]' \\ &= f'(x) \frac{1}{g(x)} + f(x) \left[-\frac{g'(x)}{[g(x)]^2}\right] \\ &= \frac{f'(x) \cancel{[g(x)]}}{\cancel{[g(x)]} g(x)} - \frac{f(x) g'(x)}{[g(x)]^2}\end{aligned}$$

Theorem (Quotient Rule) Let f and g be differentiable fns w/ $g(x) \neq 0$. Then

"high" $\rightarrow$ $\left[\frac{f}{g}\right]' = \frac{f'(x) g(x) - f(x) g'(x)}{[g(x)]^2}$
"low" $\rightarrow$

"low $\downarrow$ - high

high $\downarrow$ - low

Square the low

Away we go"

$$\begin{array}{|l} \text{Ex 1} \\ f \rightarrow \\ g \rightarrow \end{array} \left| \begin{array}{l} \left[\frac{1}{x} \right]' = -\frac{1}{x^2} \\ \left[\frac{1}{x} \right]' = \frac{[1]' \cdot x - 1 \cdot [x]'}{(x)^2} \end{array} \right.$$

$$= \frac{0 - 1 \cdot 1}{x^2} = -\frac{1}{x^2}$$

NOTE: This shows the Power Rule is true for x^n where n is any integer

$$x^n = \frac{1}{x^{-n}} \text{ positive}$$

Ex 2 Find y' if $y = \frac{\sin x}{x^2 + x}$

$$f(x) = \sin x \longrightarrow f'(x) = \cos x$$

$$g(x) = x^2 + x \longrightarrow g'(x) = 2x + 1$$

$$y' = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

$$= \frac{[\sin x]'(x^2 + x) - \sin x [x^2 + x]'}{(x^2 + x)^2}$$

$$= \frac{(\cos x)(x^2 + x) - \sin x(2x + 1)}{(x^2 + x)^2}$$

Example 3

Find y' if

$$y = \frac{2e^t}{e^t - 5}$$

$$\begin{aligned}
 y' &= \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2} \\
 &= \frac{[2e^t]'(e^t-5) - 2e^t[e^t-5]'}{[e^t-5]^2} \\
 &= \frac{2e^t(e^t-5) - 2e^t(e^t)}{(e^t-5)^2} \quad \neq [e^{t^2}]' \\
 &= \frac{2e^{2t} - 10e^t - 2e^{2t}}{(e^t-5)^2} \\
 &= \frac{-10e^t}{(e^t-5)^2} = \frac{-10e^t}{e^{2t} - 10e^t + 25}
 \end{aligned}$$

Example 4 Let a be a constant (parameter)

$$\begin{aligned}
 \left[\frac{x^2 - a^2}{x-a} \right]' &= \frac{[x^2 - a^2]'(x-a) - (x^2 - a^2)[x-a]'}{(x-a)^2} \\
 &= \frac{2x(x-a) - (x^2 - a^2)(1)}{(x-a)^2} \\
 &= \frac{2x^2 - 2ax - x^2 + a^2}{(x-a)^2} \\
 &= \frac{x^2 - 2ax + a^2}{(x-a)^2} = \frac{(x-a)^2}{(x-a)^2} = 1
 \end{aligned}$$

Derivatives of Trig Functions

Recall $[\sin x]' = \cos x$ and $[\cos x]' = -\sin x$

Tangent

$$[\tan x]' = \left[\frac{\sin x}{\cos x} \right]' = \frac{[\cos x] \cos x - \sin x [-\sin x]}{\cos^2 x}$$
$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \boxed{\sec^2 x}$$

Example 5 Find y' if $y = \cos x \tan x$

$$y' = [\cos x]' \tan x + \cos x [\tan x]'$$
$$= [-\sin x \tan x + \cos x [\sec^2 x]]$$

$\leftarrow \frac{1}{\cos^2 x}$
 $\leftarrow$ you can stop here

$$= \frac{-\sin^2 x}{\cos x} + \frac{1}{\cos x} = \frac{1 - \sin^2 x}{\cos x}$$

$$\sin^2 x + \cos^2 x = 1 \Rightarrow 1 - \sin^2 x = \cos^2 x$$

$$= \frac{\cos^2 x}{\cos x} = \cos x$$

Cotangent

$$[\cot x]' = \left[\frac{\cos x}{\sin x} \right]' = \frac{(-\sin x) \sin x - \cos x (\cos x)}{\sin^2 x}$$
$$= \frac{-(\sin^2 x + \cos^2 x)}{\sin^2 x} = \frac{-1}{\sin^2 x} = \boxed{-\csc^2 x}$$

Example 6 If $y = \underbrace{e^x}_f \cdot \underbrace{\cot x}_g$

$$\begin{aligned} y' &= [e^x]' \cot x + e^x [\cot x]' \\ &= e^x \cot x + e^x [-\csc^2 x] \\ &= e^x [\cot x - \csc^2 x] \end{aligned}$$

Secant

$$\begin{aligned} [\sec x]' &= \left[\frac{\overset{f}{1}}{\underset{g}{\cos x}} \right]' = \frac{[1]' \cos x - [1] [\cos x]'}{\cos^2 x} \\ &= \frac{0 - (-\sin x)}{\cos^2 x} = \frac{\sin x}{\cos^2 x} = \frac{1}{\cos x} \cdot \frac{\sin x}{\cos x} \\ &= \boxed{\sec x \cdot \tan x} \end{aligned}$$

Example 7 Diff. $y = \sec^2 x = (\sec x) (\sec x)$

$$\begin{aligned} y' &= [\sec x]' \sec x + \sec x [\sec x]' \\ &= (\sec x \tan x) \sec x + \sec x (\sec x \tan x) \\ &= \boxed{2 (\sec x \tan x) \sec x} \leftarrow \text{Final Answer} \\ &= 2 \sec x (\sec x \tan x) \\ &= \frac{d(\sec^2 x)}{d(\sec x)} \cdot \frac{d}{dx} (\sec x) \end{aligned}$$

Cosecant

$$[\csc x]' = \left[\frac{1}{\sin x} \right]' = \frac{[1]' \sin x - [1][\sin x]'}{\sin^2 x}$$

$$= -\frac{\cos x}{\sin^2 x} = -1 \cdot \frac{1}{\sin x} \cdot \frac{\cos x}{\sin x}$$

$$= \boxed{-\csc x \cot x}$$

Example 8

$$\begin{aligned} [\underbrace{x^5}_f \csc x]' &= 5x^4 \csc x + x^5 [\csc x]' \\ &= 5x^4 \csc x + x^5 (-1 \cdot \csc x \cdot \cot x) \\ &= x^4 \csc x (5 - x \cot x) \end{aligned}$$

Example 9 Find $h'(7)$ if $h(x) = \frac{x^2 - 14}{10 - x}$

$$h'(7) = \frac{f'(7)g(7) - f(7)g'(7)}{[g(7)]^2}$$

$$f(x) = x^2 - 14 \rightarrow f'(x) = 2x$$

$$g(x) = 10 - x \rightarrow g'(x) = -1$$

$$\begin{aligned} f(7) &= 25 & g(7) &= 3 \\ f'(7) &= 14 & [g(7)]^2 &= 9 \\ & & g'(7) &= -1 \end{aligned}$$

Editor Note
Should be 35

$$h'(7) = \frac{f'(7)g(7) - f(7)g'(7)}{[g(7)]^2}$$

$$= \frac{14 \cdot 3 - 25(-1)}{9} =$$

Editor Note:
Arithmetic error

$$\frac{45 + 25}{9} = \frac{70}{9}$$